DESIGN FOR IMPACT (D4i): A FRAMEWORK FOR TEACHING SUSTAINABILITY IN ENGINEERING DESIGN

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ABSTRACT

Sustainability has become an integrative part of engineering education since it is not possible to discuss sustainable development without also talking about innovation capability. Political and environmental frameworks request for a drastic change in the industrial landscape and also in the way design is carried out. This paradigm change forces new approaches to education that align with the prospects of the industry and also embed considerations related to the Triple Bottom Line (i.e. economic, ecological, socio-cultural elements). Addressing the complexity of sustainability requires innovative practices for teaching and learning, leading to new methodologies that aim to develop the broad sets of competencies required from the students. In Engineering Design, theories and methods related to sustainability have been mainly focused on the Design for X elements, material circularity, and product lifecycle leaving behind the importance of contextualized knowledge of regulations, or human-related aspects that motivates the students to tackle these challenges. Therefore, this study proposes a holistic approach that encompasses a broader understanding of what educators can exploit for capacitating future engineers in sustainability-related complex problem-solving. The framework highlights three main areas to be considered when teaching sustainability for Design Engineers: i) Context & Resources, ii) Human factors & Competencies, and iii) The D4i design process. A simplified version of this framework in class as a lecture-workshop format are presented and discussed along with multiple directions for future research.

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1 INTRODUCTION

Undoubtedly sustainability is an integral part of Engineering Education (EE) and vice versa. Learning and teaching frameworks are instrumental to guide and reflect on teaching practice and to engage in academic conversations as well as consider strategies that integrate sustainability concepts and applied industry cases into nearly all core teaching subjects as part of active learning.

Despite the inherent urgency and general knowledge about the importance of sustainability for all societal segments, engineering students and educators might find themselves struggling to understand how to tackle such complex topics within the educational timeframes and how to effectively embed methodologies that promote the development of competency and of a lasting mindset of sustainability.

UNESCO (2002) promotes that “Education for sustainability seeks to empower people of all ages to assume responsibility for creating a sustainable future”. Therefore, reformation of the educational systems and open dialogue became necessary to prepare future engineers and managers to have the readiness to lead sustainability policies [1]. There is a need for embracing the individual capability of the engineer into creating lasting positive impact as a byproduct of personal drive and professionalism, and for extolling sustainability as a core competency to be taught and developed during engineering education.

This study proposes the development of a human-centric framework for teaching and learning sustainability in engineering education that aims at catalysing dialog between faculty, students, and industry stakeholders in order to answer the following questions:

- How can we promote an open discussion on sustainability as a core competency in design education?
- How can we integrate a holistic human-approach to sustainable design?
- How can we empower the designer as the decision-maker towards a sustainable mindset?

Therefore, we present the Design for Impact (D4i) framework for integrating sustainability within the engineering design curricula, and present initial feedback from students after the introduction of this methodology to their studies. D4i builds upon Design for Sustainability (DfS) methodologies and looks into equipping the designer with the capabilities to assess and reflect on the impact of a given product, service, or system with a positive impact for businesses by deconstructing knowledge silos via systemic thinking. D4i looks into a transversal approach to the triple bottom line of sustainability, empowering the individual designer to develop solutions that provide positive or even regenerative impact with depth and scale in product development. This approach positions the designer as a decision-maker who is responsible for administering a systemic overview of the impact and value creation for their concept developments.

This article examines the development of this teaching framework and provides insights from a study case where a short version of the D4i methodology is tested in a lecture-workshop class.

2 THEORETICAL BACKGROUND

2.1 Design for Sustainability

Over the years, several theories provided the foundations for building an understanding of how design researchers could start linking design theory and practice with sustainability transitions, in what we know today as Design for Sustainability (DfS). Models, tools and methods can be found in the literature for sustainable product and service design, but the authors recognize limitations in their implementation and in generating sustainable solutions [2]. Therefore, DfS needs to develop towards being strongly linked to the conditions and characteristics of circular economies, overlooking digital business models [2][3], where producers and consumers are an active part of a paradigm change and are aware of subjects such as closed material loops, reuse, remanufacturing, waste reduction and recycling and even of biological cycles [5].

2.2 Sustainability education in Design Engineering

The concept of “Education for Sustainable Development” has been explored over the past three decades and many educational associations focusing on sustainability have been established. In engineering education, this concept not only develops and guides the engineering curriculum to cover sustainability issues but also requires the engineering process to meet current challenges [6]. In this way, most approaches used to integrate more practices in the engineering curricula are through capstone design courses and senior graduation projects. Engineering design courses can increase the sustainability practices of future engineers by creating and implementing their designs while considering the sustainability triple-bottom-line and measuring sustainable performance indicators or other metrics. Accreditation Board for Engineering and Technology (ABET) criteria for accrediting engineering programs [7] also emphasized the need for considering sustainability as one of the outcomes related to design ability.

More than a practical skill, Goekler [6] argues that students effectively learn about sustainability when they develop the ability to think in new ways and engage with different worldviews. For students to engage with the complex challenges involved in Design for Sustainability and towards creating impactful design they need to engage in a deep cultural critique on the present ways of production and consumption, and be equipped with a set of competencies that build on the pre-requisite abilities to think, do, analyze, plan and make decisions. This wholeness of the individual character brings the need for a human-centric understanding of the designer, moving away from the exclusively technical allusion of DfS, and reflecting on what we understand as a Sustainable Mindset [8] as an integrative part of the designer’s professional identity [9] that is developed throughout their education.

2.3 The need for a new framework

Within design education, some competencies can be acquired through formal learning, while some are embedded in one’s personality. It is the educator’s role to identify these competencies and direct their pedagogical approach and curriculum towards nurturing these different aspects of competencies. There has been considerable previous research that looked at educational practices and desirable competencies in design

[10], at dedicated competencies for building assessment framework [10][1], and at various pedagogical approaches to improve the efficacy of sustainability education in engineering such as project-based learning [12], guided-discovery learning [13], and problem-based learning [14]. Research has also focused on identifying goals and strategies to consider while performing design activities that can be adapted to sustainable design. However, existing literature and training on sustainable design does not propose any specific activities [15] nor extensively explore the role of intrinsic motivations and mindsets in the development of policies in EE.

Learning and teaching frameworks are instruments that invite faculty to reflect on their teaching practice and to engage in conversations on the teaching values that define their work at an institutional level [16]. Therefore, there is need for a framework that includes human aspects of the design engineer in the equation of sustainability teaching and learning. The designer's personal drive and mindset and its reflection in terms of professional identity is an important element of agency and professionalism that dignifies sustainable solutions and entices decision-making and behavior towards sustainability. For this purpose, we have developed and assessed a new framework.

3 DESIGN FOR IMPACT (D4I) FRAMEWORK

Design for Impact (D4i) is an experimental educational framework that builds upon DfS and looks into equipping the designer with the capabilities to assess and reflect on the impact of a given product, service, or system. Sustainability is currently defined through the interconnected domains of environment, economy and society, which is defined as the triple bottom line (TBL). Much has been discussed about new ways of seeing the interrelation of the TBL approach for sustainable development, where sustainability lay in the cross-section of equal-sized components and is now understood as facets or even layers within each other. The argument for what is now described as a “strong sustainability” approach follows a logic flow where economy is a feature of human society, which is in itself embedded in the biosphere of the planet. To take it a step further, Stockholm Resilience Center [17] introduced the concept of transversality with the “wedding cake” approach to the United Nations’ Sustainable Development Goals (SDG) while talking about the impact of pervasive systems.

Inspired by these new constructs, D4i attempts to look into a transversal approach to the TBL of sustainability, and on how to empower the individual designer to develop solutions that can tackle complex real-world problems in the digital age and provide positive or even regenerative impact with depth and scale. This approach positions the designer as a decision-maker who is responsible for administering a systemic overview of the impact and value creation for their concept developments. An illustration of the development of this understanding on sustainability approaches is presented below in Figure 1.

The core aspect of D4i as a methodology is the human-centric approach to sustainable development by placing the designer as a change-maker, i.e. a professional who can contribute to the development of a fair and sustainable society through the design of digital/non-digital products, services, and systems with transversal positive impact.

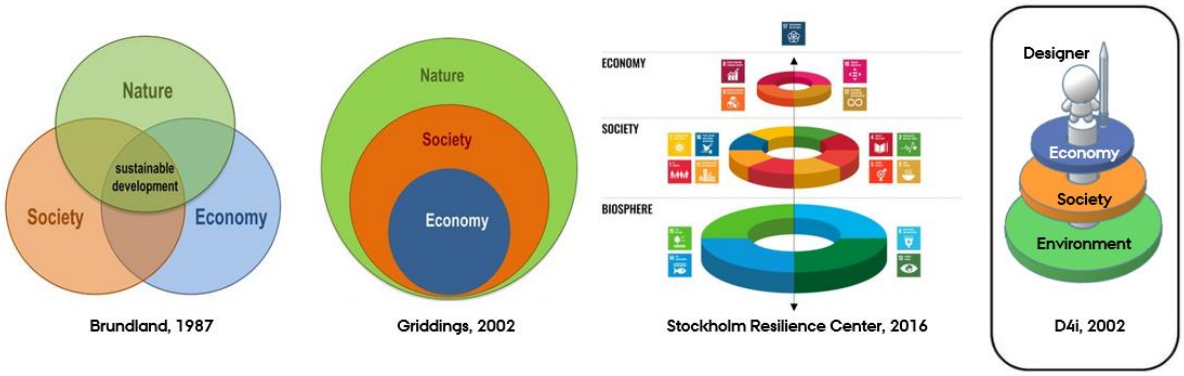


Figure 1. Evolution of sustainability models and the D4i approach

The structure of the D4i framework focuses on the need for integrating a holistic framework that relates contextual and humanistic factors with a systemic approach for technical elements as well as business understanding. The D4i framework is conceptualized as a toolkit that brings together an inventory of methodologies and techniques to facilitate a holistic understanding of the impact of a given design concept or product, aiding decision-making based on the engineers' personal drive for creating impact and business value. In this toolkit, the visual overview of D4i components tries to connect current global frameworks, human-centric aspects, and product development methodologies. D4i has three core components:

1) Context and Resources is the component that contains an assortment of existing knowledge that can guide and support the development of a new sustainable Products, Services and Systems (PSS). This component is divided into four aspects: a) Global Frameworks, which provide an overview of current assessment opportunities; b) Regulatory Framework, which provides an overview of policies, restrictions, and possibilities in relation to developing compliant solutions; c) Business and Implementations, which presents an assortment of tools and methodologies that ensure successful deployment of a solution; d) Methodologies and Resources, which provides a pool of design approaches and tools for successful engineering development. This component can be integrated in the Engineering Education (EE) as a complementary part of the traditional curricula, focusing on introducing updated frameworks and core knowledge of business along with technical methodologies. It aims at providing the students with capabilities related to systemic thinking and the overview of contextual/legal constrains. A practical example would be, in a course where the students develop new products, to present The European Union's Ecodesign Directive (Directive 2009/125/EC).

2) Human Factor is the component that contains the individual agency of the engineer. This component is divided into three aspects: e) Personal Drive, which reflects the motivation and behavioral aspect of the individual; f) Acquired Knowledge, which reflects the cognitive process and expertise of the individual; g) Professional Practice, which reflects the will for engagement with developing a solution. This component can be assessed in EE as part of a course learning outcomes, where the educator can evaluate the engagement and performance of the student. It aims to

consider individuals along with their psychological states as fundamental parts of the design process. A practical example of the implementation of this component in a course would be based on regularly monitoring the level of commitment and interest of the students, thus providing adaptations in content or approach that promotes higher engagement.

3) D4i Process is the component where all the above comes together as a design process and professional practice for the solution development and is divided into four aspects: h) Design Process; i) Impact Assessment (TBL); j) Business Potential; and k) Purpose & Impact. This component is where the individual dares to question traditional approaches by making use of critical thinking to evaluate the impact creation of their solution across the TBL and its value chains, and it can be assessed in EE as innovative approaches or outcomes. It aims at consolidating the framework's components 1 and 2 into the design process. A practical example of utilizing this component in a course would be a holistic approach to PSS development, starting with an innovative concept and ending with the communication of purpose and impact based on assessments developed as part of the design process.

Based on the components listed above, the D4i framework proposes an overview of areas of interest as means to open a dialogue on what is needed to know and to do when developing sustainable solutions. Acting as a map, the information structure of D4i framework relates contextual aspects to the PSS development, assists in systemic thinking, and includes human aspect in relation to impact creation. Figure 2 illustrates this structure and the flow of information of the D4i framework, also positioning the design engineer as the mediator between information and action.



Figure 2. Information Structure of the D4i framework

The integration and testing of this framework within an educational setup should be simplified and adapted to fit into existing courses and curricula. The D4i framework can be used as an instrument for educators to plan and develop meaningful coursework that focuses on unveiling knowledge gaps, deconstructing pre-established assumptions, and communicating impact to a given concept or philosophy such as Agile (continuous iterations for integration), Lean (continuous improvement to minimize waste) or Design thinking for co-creation with users and society. Aspects of this framework are meant to be co-developed further and updated frequently in the light of new knowledge as well as changes in context or resources.

4 CASE STUDY AND RESULTS

At this stage, the D4i framework is being developed with the purpose of assisting teaching and an open discussion in higher education setups. Therefore, we further examine the deployment of the framework and provide insights from a case study where it was tested in a lecture-workshop class for the course Advanced Product Development at Master level, from the Technology Based Business Development (TBBD) degree at Aarhus University in Denmark.

4.1 Framework adaptation for case study

In order to iterate the D4i framework, and given the time limitations for deployment in a 4h lecture, we have developed a concise version of the framework focused on specific aspects. The program was divided in two parts: i) Lecture and ii) workshop. The first half of the program focused on the D4i component of ***Context and Resources***, where the students were presented with an overview of current global actions, policies and regulations in relation to sustainability and product development. The aspect of ***methodologies and resources***, was the least exploited subject during the program due to the fact that the overall TBBD curricula covers extensively these topics. Therefore, we have only focused on introducing the concepts and briefly presenting sustainability-oriented methodologies that could serve as inspiration. The second half of the program focused on the aspect of ***Practice*** by making use of a workshop format as a space for the students to explore the ***D4i Process*** component.

The students were therefore encouraged to apply a pre-developed toolkit* into their semester-long projects. The toolkit guided the students into three sets of activities during the workshop session: 1) Concept diagnose (individually), 2) Evaluating areas of improvements and discussing within the groups, 3) Communicating impact and potential value creation. At the end of the session, the groups pitched their products in the light of their newfound understanding of impact creation. Hence, the students were able to navigate through several aspects of D4i process and reflect upon the decisions and solutions they have been developing.

4.2 Application of D4i in a Lecture-Workshop format

This case study is centered on three student groups consisting of 2-3 participants that worked together during the class using their main course assignment as a subject. This course assignment was to develop of a product-service system throughout the whole semester, where they integrated the knowledge of all the lectures provided by the curricula. The students' projects featured in the course were very distinctive between the groups, which provides D4i with a broad scope of applications. The students' highlighted the benefits of using pre-developed templates such as the toolkit for systematically navigating the complexity of the D4i framework topics. However, the limited lecture time restricted any in-depth exploration of concepts such as Life Cycle Analysis (LCA) and materials' metadata, which created uncertainty for their solutions.

* *Toolkit templates available upon request to the authors.*

After going through the exercises of the D4i toolkit, the students were encouraged to hand-draw a poster about their solutions as shown in Figure 3, and provide a 5 min pitch presentation as practice in communication of impact related to their solutions.

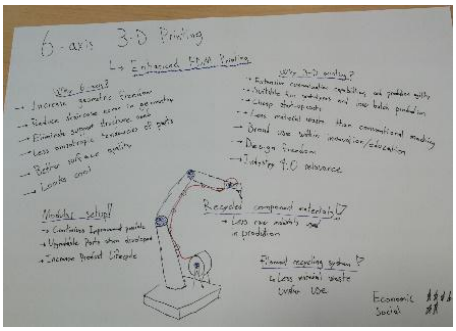


Figure 3a

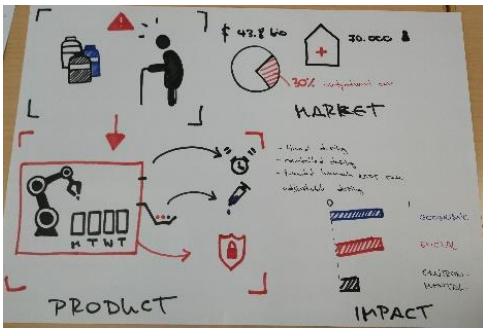


Figure 3b



Figure 3c

Figure 3. the student posters developed during the lecture in sustainable product development using the D4i framework referencing to their different group products a) a 6 axis robotic 3D printer for recycling plastic material, b) a robotic medical pill dispenser for sustainable use of medicines in elderly care dormitories, c) an industry 4.0 drone manufacturing line for repair and reuse of drones and its component materials.

The pitch and discussion session opened the opportunity for giving and receiving feedback from their peers as well as reflecting on the process of going through the exercises. The pitching practice aimed to develop communication capacity in relation to value creation across the TBL, and a space for peer discussion and support as well as to facilitate a simplified diagnosis for impact assessment as an exploratory tool for systemic thinking during product development, starting with the overlook of broad aspects related to the concept such as Product Systems Services (PSS) type, competitors and market potential, and driving to more specific design aspects such as manufacturing and components, value chain, usability and lifecycle. The students reported this class to be their first contact with a structure that allowed them to reflect on sustainability from multiple perspectives and throughout the product value chain. The toolkit and D4i framework has assisted with providing guidance and a thought flow, which inevitably aids the communication aspect of product development.

The process of assessing their previous work and evaluating impact creation in the work they developed throughout the semester has shown to provide a space for reflection and reconsideration. Among the feedback provided by the students in relation to the framework and their learnings, we observed that students present very distinct backgrounds and might not be familiar with traditional design methodologies or approaches, including basic eco-literacy, and are not familiar with the available tools for sustainable product development. As an example, despite having had previous contact with the UN SDGs, the students reported to not have a clear picture of how to integrate the goals into their concept formulations or to use the framework to reflect on the impact of their concept. Only one student had used the SDGs before as a tool to communicate value creation. By using the framework, students were able to provide clear communication on the possible impacts of their products and reflect upon the TBL aspects of their concept in relation to both manufacturing processes and usability.

5 CONCLUSIONS AND ACKNOWLEDGMENTS

This study proposes the development of D4i as a human-centric framework for teaching and learning sustainability in engineering education that aims at enticing dialog between faculty, students, and industry stakeholders in order to answer the following questions: i) How can we promote an open discussion on sustainability as a core competency in design education?; ii) How can we integrate a holistic and humanistic approach to sustainable design?, and iii) How can we empower the designer as the decision-maker towards a sustainable mindset?

The consolidation and iteration of the D4i framework shed light on the promotion of pedagogical methodologies that promote open discussion about the implementation of sustainability as a core engineering competency, and the mechanisms that support achieving this goal such as re-orienting education towards a purpose-driven practice-based-learning approach that provide tools to build long-term sustainable mindsets. Since D4i framework serves as mapping tool to identify learning gaps and promote structured reflections, educators can easily implement such framework and design their own toolkits as part of their already existing curricula and coursework, with minimal use of external resources. The limitations are mainly related to the available time during class for exploration of complex topics. However, different formats can allow for more time and for the association with multiple courses within an EE program.

This study gave us invaluable insights on the structure and usage of such a framework, approaches on how to integrate holistic elements of sustainability into restricted educational timeframes, and feedback from an initial implementation in class. Furthermore, it sheds light on new research pathways and areas that require more attention when developing useful tools for teaching, learning, and assessing competencies in the context of sustainable mindsets, professionalism in engineering, and design for sustainability approaches in education. Future work could aim to evaluating the D4i framework in a course-long setup as well as investigating what are the areas of strength and any knowledge gaps in current engineering curricula.

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